

Apex Staircase Tower

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Pre-Galvanized
Anti-Skid Checkered Plate

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IS 3696-1987

IS 4014-1967

IS 2062

EN ISO 1461

BS EN 12811-1

OSHA 29 CFR 1926.451

The **Apex Staircase Tower** is Prime Steeltech's most premium and comprehensively engineered cuplock scaffolding staircase tower, built for demanding construction and industrial environments. It features an **Anti-Skid Checkered Plate** surface on all stair treads and landing platforms, delivering strong directional grip in wet, muddy and high-traffic conditions while carrying heavy site loads. Full-height guard rails at 940 mm and mid-rails at 470 mm run along both stair sides, 150 mm toe boards protect all three open landing edges, and adjustable base jacks with 450 mm of travel handle uneven ground. Mandatory tie bars anchor the tower to the RCC structure at every level.

The 8-leg apex configuration provides maximum rigidity across a generous 3,800 mm tower width, supporting stair flights above 60 m in platform height. Compatible with all major scaffolding systems and compliant with IS, BS EN and OSHA standards, the Apex is built for high-rise, industrial and infrastructure projects.

STEEL GRADE	IS 1161
CHECKERED PLATE STANDARD	IS 3502
WELDING	MIG · IS 816
STAIR SURFACE	Anti-Skid Checkered Plate
STAIR WIDTH/ LANDING WIDTH	1 m
GRID SIZE	3800 x 2200 mm



Apex staircase tower

Overall Tower Geometry

GRID SIZE (L × W) 2,200 × 3,800 mm	PLATFORM HEIGHT Up to 60 m
TOWER OVERALL HEIGHT Platform height + 1,500 mm (top rail)	NO. OF VERTICAL MEMBERS 8 legs

Stair Geometry

Parameter	Specification
Stair Width (clear)	900 mm (min. 600 mm per EN 12811-1 Cl. 5.3; min. 457 mm per OSHA 29 CFR 1926.452(v))
Stair Type	Step arm with anti-skid Checkered plate (IS 3502)
Stair Material	M.S. Checkered plate (anti-skid) · IS 3502
Side Railing Height	940 mm
Guard Rail (Mid Rail)	470 mm
Railing - Both Sides	Yes
Step Rise	205 mm
Tread	310 mm (Class-B per EN 12811-1; BS 1139-3)
Staircase Angle	41° (EN 12811-1: 30–55°; OSHA 1926.451: 40–60°; BS 1139-3)
Step Arm UDL (Design Load)	450 kg/m ² (≥ 1.0 kN/m ² per EN 12811-1 Cl. 6.2.4; IS 3696-1)
Toe Guard - Both sides	Yes

Landing Platform — Quick Reference

LANDING WIDTH 2,200 mm	LANDING DEPTH 1,000 mm	TOE GUARDS 3 sides · 150 mm
LANDING FIXING Hook-on, secure	PLATFORM UDL 450 kg/m ²	SURFACE Anti-Skid Checkered

- Cuplock Transoms are provided underneath landing platform to avoid sagging.

Structural Framing

Component	Specification
Cuplock Vertical (Standard)	48.3 OD × M.S. IS 1161 Grade YST210
Top Cup	Drop-forged steel
Bottom Cup	Pressed metal
Cup Spacing	500 mm centers
Cuplock Ledger	48.3 OD M.S. pipe, forged ledger blades each end
Spigot Pin	38 mm OD pipe + 2 nuts & bolts (Mandatory if length greater than 250 mm)
Bracing Pipe	48.3 OD M.S. pipe, all 4 sides
Couplers & Clamps	Heavy-duty forged; IS 2750 / EN-74 compliant

Base & Anchorage

Component	Specification
Adjustable Base Jack	Spindle 38 mm OD · total length 600 mm · adjustable 450 mm · base plate 150 × 150 × 5 mm M.S.
Tie Bar	48.3 mm OD M.S. pipe + welded holding plate 150 x 150 x 6 mm M.S. · anchor fastening provision · secures tower to RCC / walls

All dimensions in mm unless noted. Weights subject to ±3.0% tolerance. Structural design required for platform heights exceeding 10 m.

Applicable Standards & Design Codes

Standard	Scope / Relevance
IS 1161:2014	Steel tubes for structural purposes — OD, wall thickness & mechanical properties (YST210/240/310) for all M.S. tubes.
IS 2750:1964	Steel scaffolding — fittings. Drop-forged and pressed couplers and clamps.
IS 3502	Steel Checkered plates — specification for anti-skid Checkered plate used on stair treads and landing platforms.
IS 3696 (Pt.1):1987	Safety code for scaffolds & ladders — stairway design, railings, toe-boards, platform protection.
IS 4014 (Pt.1):1967	Steel tubular scaffolding — materials. Component dimensions, grades, workmanship.
IS 4014 (Pt.2):1967	Steel tubular scaffolding — safety regs. Stairway access, handrails, tie-bar spacing, bracing.
BS EN 12811-1:2003	Temporary works equipment — scaffolds. Load class, stair width ≥ 600 mm, angle 30–55°.
BS 1139-3	Metal scaffolding — stairway towers. Stair geometry, Class-B rise/tread, tower configuration.
OSHA 29 CFR 1926.451	Scaffolds — construction industry (USA). Stair width min. 457 mm, angle 40–60°, fall protection > 3.05 m.
ISO 9001:2015	Quality management systems — Prime Steeltech QMS certification.

Key Features & Applications

Cuplock System Tool-free assembly.	Anti-Skid Checkered Plate (IS 3502) G.P./M.S. Checkered plate treads and landings; superior directional grip in wet, muddy and dusty site conditions.
Full Fall Protection Guard rail @ 940 mm & mid-rail @ 470 mm; both sides of stair throughout entire tower height.	Toe Guards — 3 Sides 150 mm toe boards on all open landing sides; prevents falling objects at every level.
Adjustable Base Jack 150 × 150 × 6 mm plate; 450 mm adjustable travel; levels securely on uneven ground.	Tie Bar Anchorage 48 mm OD tie bar with anchor fastening; mandatory anchorage to RCC / walls at every level.
Height — above 60 m Platform configurable to 60 m; tie bars mandatory beyond 3× least base dimension per IS 4014-2.	Platform UDL 450 kg/m² Design load 4.41 kN/m ² ; exceeds EN 12811-1 Cl. 6.2.4 minimum (1.0 kN/m ²) by 4.4×.
Design & Certification Structural drawings, load calculations, and ISO 9001 / third-party certification included.	Custom Sizes Grid, width, and height configurable to project-specific requirements.

Typical Applications

- High-rise residential & commercial construction
- Infrastructure: bridges, flyovers, metro viaducts
- Industrial plants: refineries, petrochemical, power
- Nuclear energy & defense sector structures
- Oil & gas platforms, jetty structures
- Demolition & decommissioning works
- Facade installation, cladding & glazing works
- Maintenance & renovation of existing structures
- Shipyards & dockyard construction
- Film production & event scaffolding
- Temporary public access: grandstands, events
- Aircraft maintenance hangar scaffolding

Safety Guidelines

SAFETY NOTICE — Never exceed the Safe Working Load (SWL) for the plank and span combination. Guardrails min 970 mm height and toe boards min 150 mm must be installed before access at height. Scaffold erection, alteration and dismantling by trained personnel only (IS 3696). Conduct a project-specific risk assessment before permitting personnel access.

- 1. Inspect every plank before installation — remove cracked, bent or heavily corroded units immediately.
- 2. Ensure hooks / end fittings are fully seated on ledger or transom before personnel access.
- 3. Max platform gap between adjacent planks: 25 mm. Use infill strips where required.
- 4. Overhang: min 50 mm, max 150 mm. Secure planks against uplift in exposed locations.
- 5. Do not mix plank types / sizes within the same bay without written engineering approval.
- 6. Anti-skid Checkered plate surface must be free of mud, oil, ice or loose debris at all times.
- 7. Display inspection tags (load class, date, inspector name) at each access point.
- 8. Field welding of stiffeners is NOT permitted — return damaged planks for factory repair.
- 9. Tie bars must be fixed to the RCC or structural frame at intervals per structural design. Max spacing: 4 m vertical, 6 m horizontal (IS 4014 Part 2 Cl. 12). Mandatory once height exceeds 3× least base dimension.

Notes & Disclaimers

- 1. Staircase towers must be designed, erected, and inspected by competent persons in accordance with IS 3696 Part 1, and IS 4014 Parts 1 & 2.
- 2. Adjustable base jack spindle: 38 mm OD pipe; total length 600 mm; usable / adjustable length 450 mm. Must not be extended beyond the usable length; use sole boards on soft or unpaved ground (IS 4014 Part 1).
- 3. Anti-skid Checkered plate treads must be inspected periodically; worn, deformed, or damaged plates must be replaced immediately.
- 4. Platform UDL of 450 kg/m² (4.41 kN/m²) is the design value. Site loadings must be confirmed by a structural engineer; do not exceed this value in service.
- 5. This datasheet is for information and quotation reference only. Structural calculations and certified drawings are prepared separately and are project-specific.
- 6. Tie bars must be fixed to the RCC or structural frame / walls at intervals per project-specific structural design. Maximum tie spacing: 4 m vertical, 6 m horizontal (IS 4014 Part 2 Cl. 12).

Contact & Further Information

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